Yash Rawal

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Professional Summary

Applied AI & Machine Learning Engineer serving a **1M+ user scale** by architecting enterprise agentic platforms, hierarchical RAG architectures, and automated intelligence pipelines across 120+ organizations. Specialized in Python, LLM orchestration (LangChain, LangGraph, ReAct loops), vector search, and model evaluation control planes. Active contributor to Spring AI, bridging generative AI and model fine-tuning with resilient production systems.

Skills

Languages & Backend: Python, SQL, Java, FastAPI, Spring Boot, PostgreSQL, REST APIs

AI & ML Engineering: Agentic AI, RAG & Vector Search, Multi-Agent Systems, Model Training & SFT, Context Graphs

AI & ML Frameworks: LangChain, LangGraph, Spring AI, PyTorch, vLLM, LlamaIndex, CrewAI, GROBID

Platforms & Models: Azure OpenAI, Claude, Amazon Bedrock, Google Vertex AI, Qwen, Meta Llama

Infrastructure & Security: Docker, Kubernetes, Linux, CI/CD, LLM-as-Judge, Red Teaming, PII Sanitization

Work Experience

Bigscal Technologies Pvt. Ltd.

AI/ML Engineer — Oct 2024 - Present

Remote

- Deployed a production-scale agentic platform and multi-agent systems serving **1M+ active ERP users** across 120+ enterprise tenants.
- Designed a capability contract layer to execute nested profiles inside a single ReAct loop, reducing execution overhead by **3x** and token consumption by 70%.
- Trained domain-specific **Qwen2.5-7B** models on A100 GPUs and fine-tuned E5-large-v2 embeddings using PyTorch to optimize query intent mapping.
- Developed the Prompt Lab evaluation control plane to collect reasoning trajectories and human preference annotations, establishing a scalable **Human Feedback (RLHF) Pipeline**.
- Implemented a hierarchical RAG pipeline indexing 10,000+ technical documents using GROBID, pdfplumber, and PostgreSQL/PGVector, achieving a **90%+ Recall@10** retrieval rate.
- Built production guardrails against prompt injections and data leaks, integrating runtime tool validation and **automated PII redaction**.

Bigscal Technologies Pvt. Ltd.

AI/ML Engineer Intern — Apr 2024 - Sep 2024

Surat, India

- Engineered clinical summarization pipelines for 116K+ medical records with 4-bit quantized Llama 3, and designed real-time WebSocket audio streaming for Indic ASR (Whisper, Vosk, NeMo).
- Researched audio digital signal processing (STFT, Mel-spectrograms, MFCC feature extraction via librosa/torchaudio) and CNN audio architectures (VGGish, ResNet), compiling custom Indic LJSpeech-format datasets.

Projects

Modular Agent Harness & Memory System ("Brain") | Open Source

Principal Creator & Architect of an open-source plug-and-play agentic framework designed as modular building blocks for flexible enterprise workflows, implementing a formal ReAct loop with step budgets, vector tool routing, and dynamic MCP settings caching.

Java, Spring Boot, PostgreSQL, WebSockets

Spring AI (spring-projects/spring-ai) | Open Source

Merged PR #5711 to propagate reasoningContent from AWS Bedrock and OpenAI-compatible models, and resolved Converse API prompt-caching TTL latency consistency.

Java, Spring Boot, Python

Mini Language Model (~85M Parameter Architecture) | Personal Research

Pretrained an ~85M-parameter transformer to ~1.43 validation loss on TinyStories; engineered FragmentStream chunked attention to eliminate O(T^2) VRAM overhead on commodity GPUs and custom Triton kernels with KV caching for low-latency inference.

Python, PyTorch, CUDA, Triton, Tesla P100/T4

Education

Master of Computer Applications (Artificial Intelligence & Machine Learning) - Parul University — 2022 - 2024

Certifications

Microsoft Certified: Azure AI Fundamentals (AI-900)	Microsoft
Microsoft Certified: Azure Fundamentals (AZ-900)	Microsoft